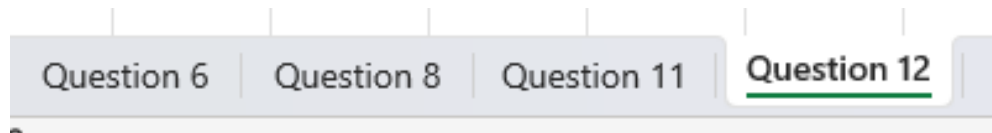


CMPG222 – Practical Assignment 1

Dear students

Please complete the following questions for Practical Assignment 1. To complete the tasks, you must study Sections 2.1 to 2.4 in the textbook. The data file is attached in Practical Assignment 1. Submit the answers to these questions as **a single Microsoft Excel document** on the eFundi assignment. Create a separate sheet for each of the four questions as follows:



Also, remember to show the sub-sections of each question clearly in each sheet, e.g.:

	A	B	C	D	E
1	Question 6 (a)				
2					
3	The answer to Question 6(a) must appear hear.				
4					
5	Question 6 (b)				
6					
7	The answer to Question 6(b) must appear hear.				
8					
9	Question 6 (c)				
10					
11	The answer to Question 6(c) must appear hear.				
12					
13					

Kind regards

The CMPG222 lecturers

Question 1

A study was conducted to analyse the weekly sales performance of sales representatives at a company. It was found that sales representatives, on average, make around 50 sales calls per week. The following data represents the number of sales calls made per week by a sample of 30 sales representatives:

52	55	62	59	49	58
48	47	48	57	53	52
54	51	56	46	55	50
60	53	50	61	60	51
49	58	45	50	47	56

- What is the least number of sales calls made in this sample? The most?
- Use a bin width of 5 calls to prepare a frequency distribution and a percent frequency distribution for the data.
- Prepare a histogram and comment on the shape of the distribution.

Question 2

A study was conducted to analyze the monthly sales (in millions of dollars) of 20 different online retail companies. The following data represents the monthly sales for these companies:

Company	Monthly Sales (Million \$)
Amazon	232.9
eBay	102.5
Walmart	75.6
Alibaba	69.8
Target	55.4
Best Buy	48.3
Macy's	44.2
Home Depot	41.6
Wayfair	39.9
Costco	38.4
JD.com	36.7
Etsy	34.2
Kroger	33.6
Overstock	32.8
Shopify	31.4
Rakuten	29.9
Zalando	28.5
ASOS	27.6
Flipkart	26.3
Snapdeal	25.1

- Which company has the highest monthly sales? Which company has the lowest monthly sales?
- Using a class width of 10 million dollars, develop a frequency distribution of the data starting with 20–29.9, 30–39.9, 40–49.9, and so on.
- Prepare a histogram of the frequency distribution. Interpret the histogram.

Question 3

A customer service department collected satisfaction scores from a sample of 20 customers. The scores range from 1 to 20, with 20 indicating the highest level of satisfaction. The following data represents the satisfaction scores collected over a month:

4, 7, 12, 15, 9, 10, 6, 16, 11, 14, 13, 12, 8, 17, 5, 14, 18, 3, 19, 8

Using classes 1–5, 6–10, 11–15, and 16–20, show the following:

- a) The frequency distribution
- b) The relative frequency distribution
- c) The cumulative frequency distribution
- d) The cumulative relative frequency distribution

Question 4

A survey was conducted to analyse the monthly household expenditures (in thousands of dollars) of 30 different households. The following data represents the monthly expenditures for these households:

3.2, 4.5, 2.8, 3.9, 5.1, 4.3, 3.0, 6.2, 3.7, 5.8, 4.1, 3.5, 4.7, 5.3, 4.6, 3.4, 5.9, 6.0, 2.7, 4.0, 3.3, 5.0, 4.8, 3.6, 4.4, 5.2, 6.1, 2.9, 5.5, 4.2

Using classes of 2.0–2.9, 3.0–3.9, 4.0–4.9, 5.0–5.9, and 6.0–6.9, show the following:

- a) The frequency distribution
- b) The relative frequency distribution
- c) The cumulative frequency distribution
- d) The cumulative relative frequency distribution
- e) What do these distributions tell you about the monthly household expenditures?
- f) Show a histogram of the data. Comment on the shape of the distribution.
- g) What is the largest household expenditure?